UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	COMPUTER ORGANIZATION AND ARCHITECTURE
COURSE CODE	CSC429
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of two (2) parts: PART A (10 Questions) PART B (1 Question)
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 7 printed pages.

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PART A (80 MARKS)

Answer ALL questions in this part.

QUESTION A1 **(5 marks)**

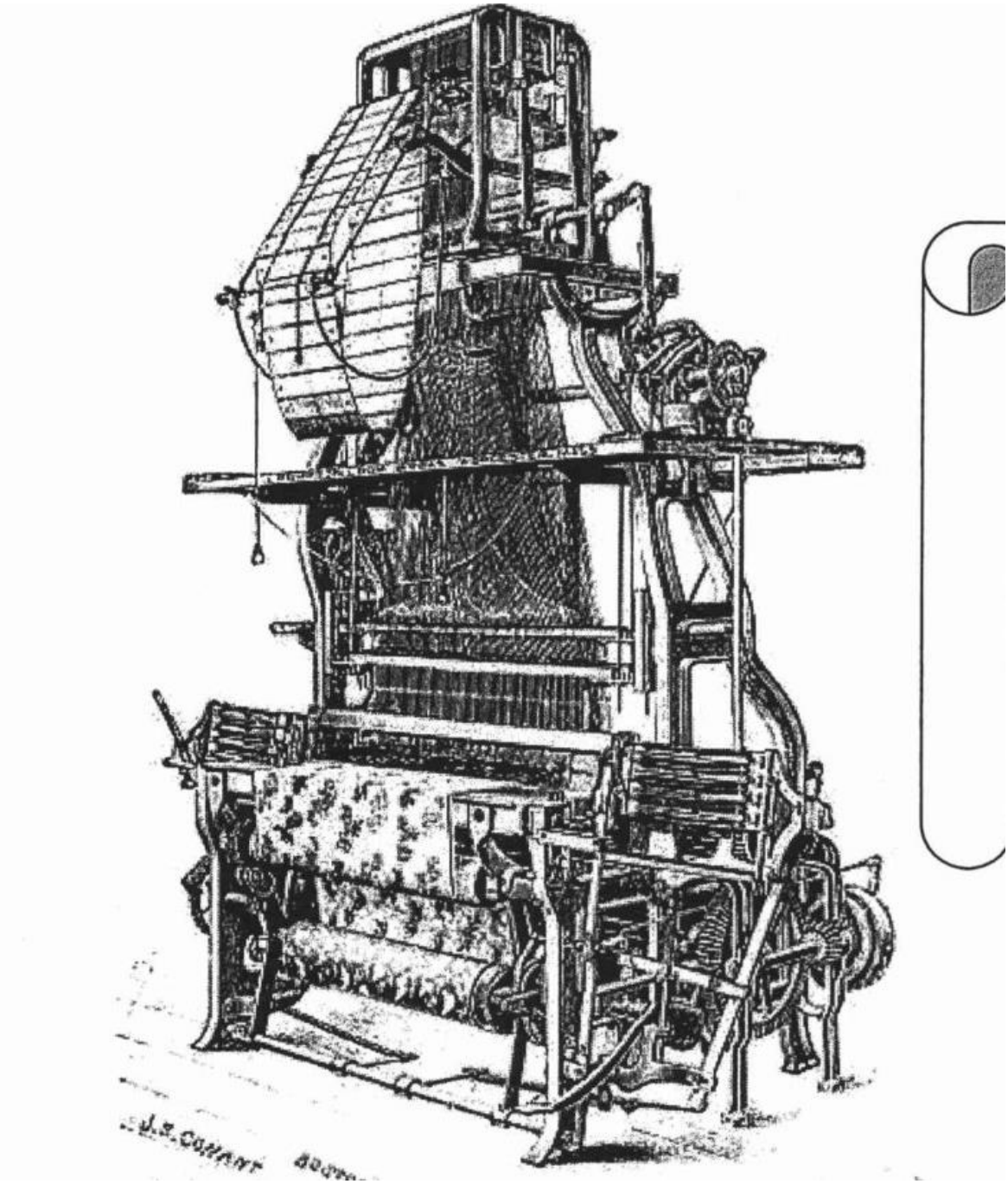
a) How classification of parallel processor is constructed?

b) Give ONE (1) example for each parallel processor architecture.

QUESTION A2 **(5 marks)**

Question 1 is based on the following placard in the Computer History Museum.

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The Jacquard System was developed by Joseph Marie Jacquard to automatically produce the intricate woven patterns. The deck of punched cards that controls the Machine X is located at the left center on the far side of the figure. In the early days of software piracy, the intricate fabric design card decks would be stolen by competing textile mills.

a) Name the Machine X in Figure 1.

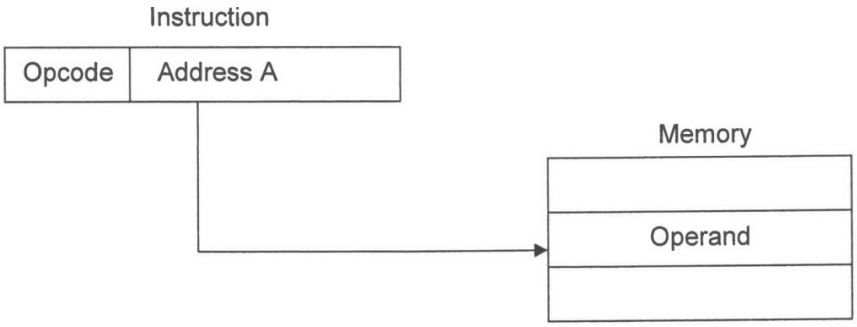
b) In which computer generation was Machine X developed?

c) Name the possible computer component that has similar function as the deck of punch card in Machine X.

d) Identify the input and output for Machine X.

QUESTION A3 **(5 marks)**

Given the following Figure 2.



a) Label the addressing technique illustrated in Figure 2.

b) Explain how the addressing technique in 9(a) is used.

List ONE (1) advantage and ONE (1) disadvantage of the addressing technique in

QUESTION A4 **(5 marks)**

Convert 3.25 to IEEE 754 single-precision format. Give the answer in hexadecimal and show the sign, exponent and mantissa fields.

QUESTION A5 **(10 marks)**

a) Explain the problem in polling 1/0 method.

b) Explain how buffering can help to improve the device overall performance?

c) Discuss how DMA technique can help the 1/0 transfer performance instead of using polling method and interrupt 1/0 method?

QUESTION A6 **(10 marks)**

Answer both parts for the same expression. Assume an 8-bit byte.

Expression: 63 - (-41)

a) Calculate 63 - (-41) using sign and magnitude.

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b) Calculate 63 - (-41) using 1's and 2's complement.

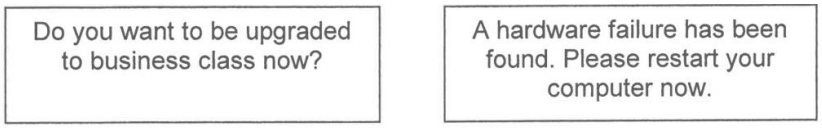
QUESTION A7 **(10 marks)**

a) Discuss the function of each Central Processing Unit (CPU) component'

b) Elaborate on what will happen in the machine cycle if the bus system is faulty

QUESTION A8 **(10 marks)**

Question 5 is based on the following scenario. "Nurathia Farhana browses the UiTM4fly.com website to book for a return flight from KLIA to LAX. While comparing for a better price, two messages popped up in her screen. Figure 1 (a) and (b) illustrate the conveyed messages.



(a) (b)

(b)

(a)

a) Which message is the non-maskable interrupt? Justify your answer by giving appropriate reason.

b) Differentiate between maskable and non-maskable interrupt?

c) Outline the common way to handle multiple interrupts as happened in the given scenario.

QUESTION A9 **(10 marks)**

a) Compare system interconnection and CPU interconnection in terms of:

i. function

ii. physical appearance.

b) Describe the function of the THREE (3) different bus types in a computer system.

QUESTION A10 **(10 marks)**

a) Discuss the main function of cache memory.

b) How can the data stored in the cache and main memory be synchronized.

c) Contrast between SRAM and DRAM based on the following characteristics: Type of storage Availability of charges leak price Example of location of memory usage

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PART B (20 MARKS)

Answer the question in this part.

QUESTION B1 **(20 marks)**

Question 1 is based on Figure 5 and Table 1

$$z = A + C^2 / B * D - 3A$$

Write an appropriate program using conect assembly language notations to compute the equation shown in Figure 5 using instructions for zero-address, one-address, and three- address. The available instructions are provided in Table 1. The registers labeled A, B, C, and D are fixed, while registers x, y, and z can be used temporarily. The final answer must be stored in register z.

Zero-address	One-address	Three-address
PUSH p	LOAD p	ADD (p,q,r)
POP p	STOR p	SUB (p,q,r)
ADD	ADD p	MUL (p,q,r)
SUB	SUB p	DIV (p,q,r)
MUL	MUL p	
DIV	DIV p	